

## Nepriylsin

*zinc metallopeptidase*

Catalytic Zn(II) held by  
His583, His587, Glu646

**Requires an ANIONIC group  
(carboxylate, thiol, hydroxamate)**

**One molecule**  
must be acidic  
AND basic

## DPP-IV

*serine protease*

Glu205 / Glu206 dyad  
in the S2 subsite

**Requires a BASIC nitrogen  
(primary or secondary amine)**

**Consequence: the molecule is a zwitterion at physiological pH.**

High polarity and low membrane permeability follow directly.

Sacubitril is administered as an ester prodrug for exactly this reason.